GOAL

The goal of each player is to reach a certain amount of points of carbon, nitrogen, and phosphorus as fast as possible (adapted from the original card game) while managing these same points for survival.

CARD ARCHETYPES

1) **ORGANISMS** - Specifically plants, algae, detritivores (shredders, collectors, grazers, etc.), and consumers/predators. These cards are to consume or to be consumed but also the main source of nutrients. Can die to insufficient nutrients, and also has evolution/adaptation mechanics.

    a. **PRODUCERS**: Plant, Algae, Etc.
    Consumes nutrients— mainly carbon and nitrogen to produce oxygen (O_2). Has a fixed amount of lifespan / rounds. Can be eaten by consumer cards played by any player.

    b. **DETRITIVORES**: Collectors, Grazers, Shredders
    Eats dead organic matter (regardless from which player), releasing small amounts of nutrients. Can not fully consume a dead card in one round.

    c. **CONSUMER**: Primary, Secondary, Tertiary Consumers
    Separated into three trophic levels. When consuming organisms below their trophic level, the nutrient pool of the organism transfers to the consumer. Their nutrient pool will be larger, allowing more nutrients available to be released when decomposing. Additionally, it can be used to protect and attack against enemy cards.

2) **MICROBE** - Bacteria/Fungi that can target all or a specific nutrient. Allows further decomposition done by detritivores, generating more nutrients to be recycled back for use.

3) **WILD EVENT** - Random events that will occur to both players when drawn. Results can be advantageous or not to both players. When drawn, the event will immediately be played but will not consume any action points.

4) **EVOLUTION / ADAPTATION** - Enhances the abilities of organisms. Must complete required conditions for activation.

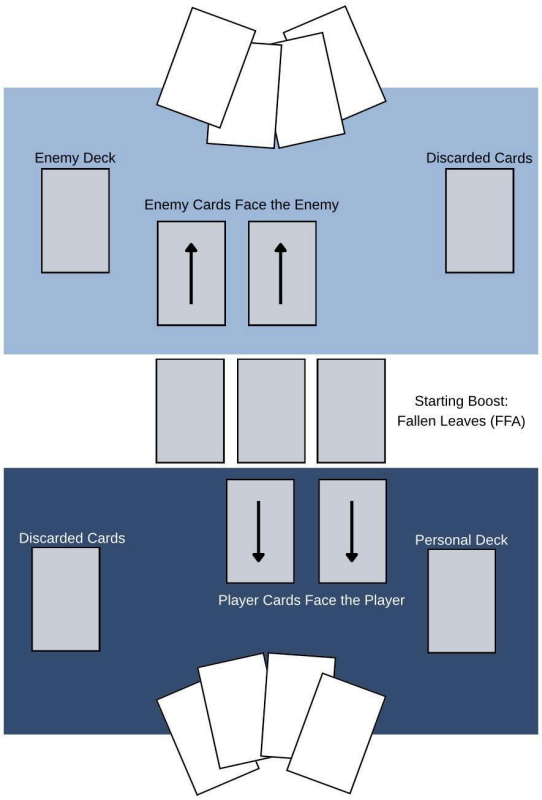
RULES & MECHANICS

(A) Player Deck, Hand, Rounds & Turns

- A deck containing # of Organism Cards, # of Microbe Cards, # of Evolution Cards, and 4 Wild Event Cards are shuffled into separated decks for each players
- Each player will start with a hand of 5 random cards
- A player has 3 action points (fixed) per round
- A card can be disposed of for a new card, costing 1 point. Cards may also be held for future rounds
- Upon placement of any cards (except wild event cards), they consume 1 action point.
- When cards are placed/played, it CAN NOT be retracted

(B) Game Set-up

- (subject to change):



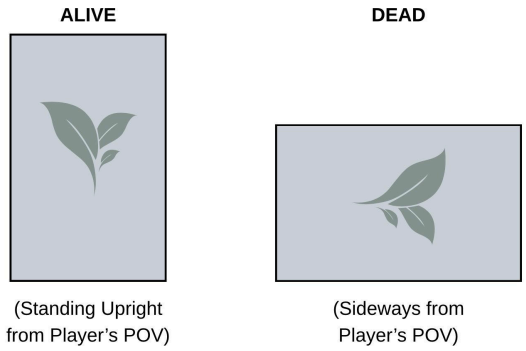
(C) Card-Specific Mechanics

(C.1.) Organism Cards have 2 statuses: Alive & Dead

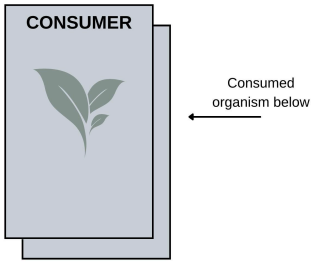
Alive - If activation conditions are satisfied, organisms will do specific processes. They may either die to the lack of necessary nutrients points or to natural causes via a number of rounds or wild event cards.

Dead / **Decaying Matter** - Organisms that can be readily consumed by detritivore cards. Using a detritivore card will separately cost another action point in the current round but will not in the succeeding rounds.

(C.2.) Placement indication for the organism cards:

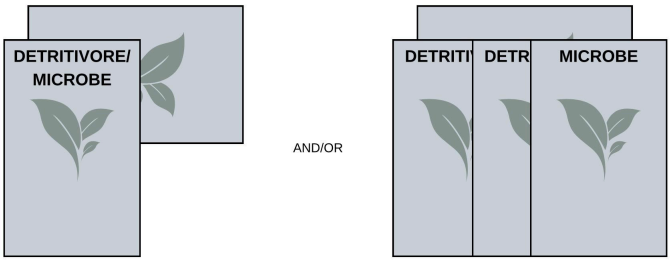


(C.3.) Consumer-Specific Interactions— indications:



- A consumed organism stays below the consumer card but stays visible for tracking purposes.

(C.4.) Detritivore and Microbe-Specific Interactions— indication:



- The decaying matter getting broken down stays below any microbe or detritivore card. It must stay visible even with stacked cards for tracking purposes.

(D) Day/Night Cycle

Main Idea : Photosynthesis at day, accumulated CO_2 at night aka respiration

Medj eme q pa toh hindi pa sure

- Photosynthesis of plant is enabled during the day, giving the player additional oxygen for more actions. This is immediately disabled during a round under the night.
- Rounds alternate between day and night, with round 1 starting at day.

(E) Point Management System

- There are 4 kinds of nutrient points to be used and earned to the player's advantage. Three (3)— carbon, nitrogen, and phosphorus— are necessarily needed to win. One (1)— oxygen is essential for managing your cards.
- The player is given starting nutrients:
    - 10 Carbon
    - 3 Nitrogen
    - 0 Phosphorus
    - 7 Oxygen
- Oxygen is not needed to win however it is used for the majority of the abilities.

(F) Winning Conditions

- A player must achieve:
    - 80 Carbon Points
    - 15 Nitrogen Points
    - 2 Phosphorus Points
- A sheet for tracking these points will be given beforehand
- Subject to change yung amount of points. Nakabase lng sha sa nakita q online, also similar ung amount don sa og card game ni zer

40-card deck
5-card starting hand
C/N/P resources
O_2 ecosystem resource
3 AP
Day/Night
Alive → Dead → Decomposing
Food chain interactions
Microbial decomposition
Evolution/specialization
Wild events
50 C / 15 N / 2 P victory

PRODUCER CARDS

Plant Card	C Cost	N Cost	O_2 Produced	Lifespan	Special Effect
Leaf Litter Plant	1	1	+2	3 rounds	When it dies, creates 1 Fallen Leaf
River Grass	2	1	+2	4 rounds	+1 C when successfully photosynthesizing
Aquatic Plant	2	1	+3	4 rounds	Requires water; unaffected by Drought once
Riparian Tree	3	2	+3	5 rounds	Cannot be consumed by Primary Consumers
Fast-Growing Plant	1	1	+1	2 rounds	Gains +1 C when played

Algae Card	C Cost	N Cost	O_2 Produced	Lifespan	Special Effect
Green Algae	1	1	+2	2 rounds	+1 C when photosynthesizing

Diatom	1	1	+2	3 rounds	When decomposed, returns +1 P
Bloom Algae	2	1	+3	2 rounds	If P ≥ 3, triggers Algal Bloom
Green Algae	1	1	+2	2 rounds	+1 C when photosynthesizing

SPECIAL PRODUCER MECHANICS

- A normal producer that has been upgraded using an Adaptation card

Example:

River Grass (normal stats and abilities)

- During Action Phase -

Play Adaptation Card

Adaptation Card

- Requires an existing producer
- Costs 1 AP
- Costs 2 C + 1 N

River Grass —-> Special River Grass

Place the Adaptation card underneath it to show that it is now Special.

Special Producer	Trait	Why it matters
Nutrient-Rich	When consumed, transfers **+1 C and +1 N** to the consumer	Connects producer biomass → consumer
Oxygen-Rich	Produces **+2 O_2** during the Day	Supports ecosystem oxygen balance
Phosphorus-Accumulator	Stores **+1 P** when it photosynthesizes; P returns when decomposed	Creates a clear **P cycle**

CONSUMER CARDS

Basic rule

Consumer have a **Trophic Level (TL):**

- **TL 2 – Primary Consumers:** eats producers
- **TL 3 – Secondary Consumers:** eats primary consumer
- **TL 4 – Tertiary Consumers:** eats secondary consumer

Primary Consumers

Card	C Cost	N Cost	O_2/round	Lifespan	Can consume
Caterpillar	2	1	1	3	Plants
Zooplankton	1	1	1	3	Algae
Snail	2	1	1	4	Algae / Aquatic Plants
Grasshopper	2	1	1	3	Plants

Secondary Consumers

Card	C Cost	N Cost	O_2/round	Lifespan	Can consume
Small Fish	3	1	1	4	Zooplankton / Snails
Frog	3	2	1	4	Caterpillars / Grasshoppers
Predatory Insect	2	2	1	3	Caterpillars / Grasshoppers

Tertiary Consumers

- Can consume higher-level organisms but have higher costs and higher oxygen requirements.
- Can contribute greatly to the nutrient pool when decomposed by returning C/N/P consumed.
- Cannot directly consume producers.

Card	C Cost	N Cost	O_2/round	Lifespan	Can consume
Large Fish	4	2	2	5	Small Fish / Frog
Water Snake	4	2	2	5	Small Fish / Frog
Heron	5	3	2	5	Fish / Frogs

CONSUMER EVOLUTION MECHANICS

Caterpillar consumes opponents nutrient-rich producer (special)
—> Caterpillar gains +1 specialization mark
—> Consumes another Special Producer +1 specialization mark

– Caterpillar Evolution Requirements –

- 2 Specialization Marks
- Required C/N
- Still alive at the end of the round
- 1 AP

—> Specialized Caterpillar (Consumer)
—> Higher-level & specialized card

DETRITIVORE CARDS

Shredders —> Break large dead materials
Collectors —> Consume fine organic matters
Grazers —> Consume algae/biofilm

Shredder Cards	C Cost	N Cost	O_2/round	Lifespan	Ability
Freshwater Shrimp	2	1	1	4	**Shred:** Consume 1 Dead Plant. Place a Shredded Detritus marker on it. The next Microbe that decomposes this detritus gets +1 C/N returned.
Aquatic Worm	1	1	1	3	**Burrow:** When processing Dead Detritus, reduce the decomposition time by 1 round.

Collector Cards	C Cost	N Cost	O_2/round	Lifespan	Ability
Tubifex Worm	1	1	1	3	**Filter:** When a Dead organism is decomposing, gain +1 C when collecting its detritus.
Chrinomid Larva	2	1	1	3	**Fine Detritus:** Can process detritus that has already been shredded or partially decomposed. Gain +1 N if it does.

Grazer Cards	C Cost	N Cost	O_2/round	Lifespan	Ability
Freshwater Snail	2	1	1	4	**Graze:** Consume 1 Algae/Biofilm token → gain +1 C. Can prevent algal bloom.

Aquatic Grazer	1	1	1	3	**Control Algae:** When consuming algae, remove 1 excess P associated with that algae. Manages phosphorus level before algal bloom becomes a problem.

MICROBE CARDS

- Return nutrients from dead organic matter to the nutrient pool.

Card	Cost	Main ability
Decomposer Fungus ×2	1 AP	Decomposes 1 processed dead organism and returns its stored C/N/P to the nutrient pool.
Nitrogen Bacteria x2	1 AP	When decomposing Processed Detritus, return its N and gain +1 N.
Phosphorus Bacteria x2	1 AP	When decomposing Processed Detritus, return its P and gain +1 P.
Super Decomposer x1	1 AP	Decompose 1 Processed Detritus and return its C/N/P. This decomposition does not require a previous detritivore action.

Bacterial bonus +1N/P only once per card per game

Evolution and Adaptation Cards

PRODUCER-SPECIFIC			
Card	**AP**	**Requirement**	**Effect**
Nutrient-Rich Adaptation	1	Existing Producer + 2 C + 1 N	Producer becomes ⭐ Nutrient-Rich **Specialized Producer Effect:** Consumer gains +1 C and +1 N
Oxygen-Rich Adaptation	1	Existing Producer + 2 C + 1 N	Producer becomes ⭐ Oxygen-Rich **Specialized Producer Effect:** During the day, produces +2 additional O_2.
Phosphorus-Accumulator Adaptation	1	Existing Producer + 2 C + 1 N	Producer becomes ⭐ P-Accumulator **Specialized Producer Effect:** During photosynthesis, stores +1 in its biomass.
CONSUMER-SPECIFIC			
Consumer Specialization	1	Consumer has 2 specialization marks + meet its C/N requirements + still alive	Upgrade the consumer to its designated Specialized Consumer.
Protected Adaptation	1	Any living organism + 2 C + 1 N	The organism cannot be consumed by higher levels. Not immortal.

Wild Event Cards

Wild Event	Trigger / Condition	Effect	Ecological Concept
Strong Rainfall	Played during Environmental Phase	All players gain **+2 P**. If a player has **≥3 P** and at least 1 Algae/Bloom Algae, trigger **Algal Bloom**. During the next Night, all organisms use **+1 O_2**.	Nutrient runoff → eutrophication → algal bloom → oxygen depletion
Drought	Played during Environmental Phase	During the next Day, Producers produce **−1 O_2**. Aquatic organisms cannot be played. Aquatic Plants cannot photosynthesize.	Water limitation → reduced productivity and habitat availability
Heat Wave	Played during Environmental Phase	During the next Night, all organisms consume **+1 O_2**. Organisms that cannot meet their O_2 requirement die.	Higher temperature → increased oxygen stress / lower oxygen availability
Flood	Played during Environmental Phase	Each player chooses **1 organism**. That organism becomes **Processed Detritus**. Player gains **+1 C**. If no organism is available, gain **+1 P** instead.	Flooding → physical disturbance + transport of organic matter and nutrients